Kingsley · J Shackett

16
+ 8
14
13
51

UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2024/2025 **FIRST SEMESTER EXAMINATION**

COURSE: **MTH315: REAL ANALYSIS II**
DATE SCHEDULED: **THURSDAY 30th JANUARY, 2025**
TIME ALLOWED: **$2\frac{1}{2}$ HRS**

Instructions: Attempt any four (4) questions only.

1. (a) Classify whether the following sequence are bounded (i) $u_n = \{\frac{1}{n}\}_{n=1}^{\infty}$ (ii) $u_n = \{\frac{(-1)^n}{n}\}_{n=1}^{\infty}$ (iii) $u_n = \{(-1)^n\}_{n=1}^{\infty}$.
   (b) Determine the limit of the following sequences (i) $u_n = \{\frac{(n-1)(2n-1)(n+1)}{n^3}\}_{n=1}^{\infty}$ (ii) $u_n = \{\frac{3^{n+2}+5^{n+2}}{3^n-5^n}\}_{n=1}^{\infty}$ (iii) $u_n = \{\sqrt[n]{1+a^n}\}_{n=1}^{\infty}$.
   (c) Let $f(x) = \frac{1}{2}x^3$ and $P = \{0,1,2,3,5\}$, $Q = \{0,1,2,3,4,5\}$, $[a,b] = [0,5]$. Compute the following: (i) $U(f,P)$ (ii) $L(f,P)$ (iii) $U(f,Q)$ (iv) $L(f,Q)$. Shows that Q is finer than P.

2. (a) (a) State and prove Rolle's Theorem.
   (b) Verify Rolle's Theorem. when $f(x) = x^2+1, 1 \le x \le 4$.
   (b) State five properties of $R-S$ integral functions.
   (c) Prove that if f is monotone on the bounded closed interval $[a,b]$, then f is integrable on $[a,b]$.

3. (a) State and prove Mean Value Theorem.
   (b) Verify Mean Value Theorem when $f(x) = x^2-2x+1, -1 \le x \le 3$.
   (c) State and prove Cauchy's Mean Value Theorem.

4. (a) Define on the following terms: (i) Series $\sum_{i=1}^{\infty} a_i$ of real numbers (ii) Convergence of Series $\sum_{i=1}^{\infty} a_i$ (iii) divergence of Series $\sum_{i=1}^{\infty} a_i$.
   (b) Show that the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$, $p > 0$, converges for $p > 1$ and diverges for $p \le 1$.
   (c) Show that the series $\sum_{n=2}^{\infty} \frac{1}{n \log n (\log\log n)^p}$, $p > 0$, converges for $p > 1$ and diverges for $p \le 1$.

5. (a) State the Limit Ccomparison Test, D'Aliembert's Ratio Test and the Cauchy's Root Test.
   (b) By using Limit Ccomparison Test or otherwise, discuss which of the following converge or diverge (i) $\sum_{n=1}^{\infty} \frac{n}{n^4-3}$ (ii) $\sum_{n=1}^{\infty} \frac{n^2}{n^4-3}$ (iii) $\sum_{n=1}^{\infty} \frac{n}{e^{n^2}}$.
   (c) By using the D'Aliembert's Ratio Test or the Cauchy's Root Test or otherwise, determine which of the following series converge and which diverge (i) $\sum_{n=1}^{\infty} \frac{10^n}{n}$ (ii) $\sum_{n=1}^{\infty} \frac{n}{2^n}$ (iii) $\sum_{n=1}^{\infty} \frac{\log n}{2^n n}$.

6. (a) Define the following terms: (i) pointwise convergence (ii) uniform convergence
   (b) State and prove Cauchy-Condition for uniform convergence.
   (c) Let f_n be defined on $[-1,1]$ by $f_n = \frac{x}{1+n^2x^2}$. Show that $\{f_n\}$ converges uniformlly and the limit f is differentiable.

$\sum_{n=1}^{\infty} \frac{\log n}{2^n n}$

Limit co.. $\frac{\log n}{2n} = b_n$

Limit comp. + est $\frac{\log n}{2^n n}$ / $\frac{\log n}{2n}$

$\frac{2n \cancel{\log n}}{2^n n \cancel{\log n}}$

$\frac{2n}{2n} = \frac{1}{\infty} =$

CamScanner